

# Floorplan Generation with Graph Beta Diffusion

Yusuke Takeuchi

Tetraz, Inc.  
Tokyo, Japan  
University of Tokyo  
Tokyo, Japan  
takeuchi-yusuke122@g.ecc.u-  
tokyo.ac.jp

Qi An

University of Tokyo  
Tokyo, Japan  
anqi@robot.t.u-tokyo.ac.jp

Atsushi Yamashita

University of Tokyo  
Tokyo, Japan  
yamashita@robot.t.u-tokyo.ac.jp



Figure 1: From a beta-noised graph to a structured floorplan, then rendered as layouts.

## Abstract

We present a single-stage graph beta diffusion model for residential floorplan generation. In contrast to multi-stage pipelines, our formulation is end-to-end, avoiding error accumulation across modules and reducing the number of model parameters and hyperparameters. We further introduce an unsupervised Manhattan alignment loss that encourages axis-aligned walls. Our method substantially improves Fréchet Inception Distance over a parameter-efficient GAN baseline.

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## 1 INTRODUCTION

Architectural floorplan design is labor-intensive, especially in the early stages that require exploring many layout alternatives. Generative models can automate this process, improving efficiency and providing diverse, well-structured options that enhance communication and client satisfaction.

Since the RPLAN dataset [Wu et al. 2019], deep learning has become central to data-driven floorplan generation. GSDiff [Hu et al. 2025] models layouts as graphs but uses a two-stage node-edge pipeline, where upstream errors propagate and multiple models enlarge both parameter count and hyperparameter space.

We propose a single-stage beta diffusion framework for structural graphs, jointly generating room types, connectivity, and geometry. Building on Graph Beta Diffusion (GBD) [Liu et al. 2024],

our method simplifies the pipeline and introduces an unsupervised Manhattan alignment loss encouraging axis-aligned layouts. Our contributions are: (i) a single-model end-to-end framework, (ii) the first application of GBD to real-world structural data, and (iii) a reconstruction-free unsupervised Manhattan alignment loss that promotes axis-aligned geometry.

## 2 METHOD

### 2.1 Floorplan Representation

A floorplan is modeled as an undirected graph  $G = (V, A)$  with  $N$  junction nodes following [Hu et al. 2025]. Each node  $v_i = (x_i, y_i, r_i)$  encodes normalized coordinates  $(x_i, y_i) \in [0, 1]^2$  and a multi-hot room-type vector  $r_i \in \{0, 1\}^C$  over  $C$  room categories. Wall segments are represented by a binary adjacency matrix  $A \in \{0, 1\}^{N \times N}$  with  $A_{ij} = 1$  if nodes  $i$  and  $j$  are connected by a wall.

### 2.2 Beta Diffusion

Let  $G_0 = (V_0, A_0)$  denote the clean graph. The forward process applies element-wise Beta noise parameterized by  $G_0$ ; its closed-form marginal is

$$q(G_t | G_0) = \text{Beta}(\eta \alpha_t G_0, \eta(1 - \alpha_t G_0)), \quad (1)$$

where  $\alpha_t \in (0, 1]$  decreases as  $t$  increases and  $\eta > 0$  controls concentration (large  $\eta$  lowers variance).

The reverse process time-reverses (1) and learns per-step denoising transitions. The one-step reverse posterior is

$$q(G_{t-1} | G_t, G_0) = (1 - G_t)^{-1} \text{Beta}(u_t; a_t, b_t), \quad (2)$$

where

$$u_t = \frac{G_{t-1} - G_t}{1 - G_t}, \quad (3)$$

$$a_t = \eta(\alpha_{t-1} - \alpha_t)G_0, \quad b_t = \eta(1 - \alpha_{t-1}G_0). \quad (4)$$

At inference we predict  $\hat{G}_0 = f_\theta(G_t, t)$  and define

$$p_\theta(G_{t-1} | G_t) = q(G_{t-1} | G_t, \hat{G}_0). \quad (5)$$

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## 2.3 Training Loss

We minimize the sum of a diffusion surrogate loss and the proposed Manhattan alignment loss:

$$\mathcal{L} = \mathcal{L}_{\text{GBD}} + \lambda_{\text{man}} \mathcal{L}_{\text{man}}^{\text{scheduled}}. \quad (6)$$

*Manhattan alignment loss.* To promote orthogonal geometry, we impose an unsupervised alignment loss that penalizes off-axis edges:

$$\mathcal{L}_{\text{man}} = \frac{1}{\sum_{i,j} \hat{E}_{ij}} \sum_{i,j} \hat{E}_{ij} \log \left( 1 + \frac{\min\{|x_i - x_j|^2, |y_i - y_j|^2\}}{\sigma^2} \right), \quad (7)$$

where  $\hat{E}_{ij} = \text{sigmoid}(\kappa(E_{ij} - \tau))$  is a differentiable edge derived from the soft adjacency  $E$ . Here,  $\tau$  is a confidence threshold,  $\kappa$  controls the steepness, and  $\sigma$  sets the tolerance for near-horizontal/vertical alignment. We apply noise-aware scheduling,

$$\mathcal{L}_{\text{man}}^{\text{scheduled}} = \sum_{t=2}^T \alpha_t^\gamma \mathcal{L}_{\text{man}}^{(t)}, \quad (8)$$

where  $\mathcal{L}_{\text{man}}^{(t)}$  is computed on the step- $t$  predictions,  $T$  is the total number of diffusion steps, and  $\gamma > 0$  controls the schedule strength. This keeps the penalty small under high noise and larger at lower noise, encouraging axis-aligned walls when edges are reliable.

*GBD surrogate loss.* Following GBD [Liu et al. 2024], we use a two-term KL surrogate that upper-bounds the divergence between the learned reverse chain and the forward-defined target with trade-off weight  $\omega$ .

$$\mathcal{L}_{\text{GBD}} = \sum_{t=2}^T \left( (1 - \omega) \mathcal{L}_{\text{sampling}}(t) + \omega \mathcal{L}_{\text{correction}}(t) \right), \quad (9)$$

$$\mathcal{L}_{\text{sampling}}(t) = \mathbb{E}_{q(G_t, G_0)} \left[ \text{KL}(p_\theta(G_{t-1} | G_t) \| q(G_{t-1} | G_t, G_0)) \right], \quad (10)$$

$$\mathcal{L}_{\text{correction}}(t) = \mathbb{E}_{q(G_t, G_0)} \left[ \text{KL}(q_t(\cdot | \hat{G}_0(G_t, t)) \| q_t(\cdot | G_0)) \right]. \quad (11)$$

## 3 EXPERIMENTS

### 3.1 Implementation and Data

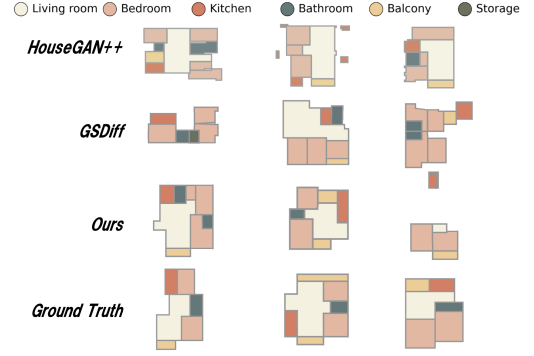
We implement the model in PyTorch with a 6-layer Graph Transformer and a Beta-diffusion schedule of  $T=1000$  steps. The forward Beta uses concentration  $\eta=10,000$  for active nodes/edges and 10 for background. Training runs for 1M steps on a single RTX 3090 using AdamW with  $lr = 10^{-3}$  and batch size is 128. Loss hyperparameters are  $\lambda_{\text{man}}=20$ ,  $\sigma=0.01$ ,  $\kappa=30$ ,  $\tau=0.9$ ,  $\gamma=3$ , and  $\omega=0.01$ . We evaluate on RPLAN [Wu et al. 2019] (~80k floorplans), following GSDiff for preprocessing to structural graphs and for the train/validation/test split.

### 3.2 Evaluation Metrics

We evaluate generation quality using the Fréchet Inception Distance (FID), which measures distributional similarity between the test set and generated floorplans. As baselines we include HouseGAN++ [Nauata et al. 2021], a parameter efficient GAN whose inference uses a single generator and thus matches our small model setting, and GSDiff [Hu et al. 2025]. We also report an ablation that removes the proposed unsupervised Manhattan alignment loss.

**Table 1: Quantitative comparison on RPLAN.**

| Method                          | FID↓   | #Param (M) |
|---------------------------------|--------|------------|
| HouseGAN++ [Nauata et al. 2021] | 48.40  | 2          |
| GSDiff [Hu et al. 2025]         | 4.83   | 125        |
| Ours (with Manhattan Loss)      | 19.49  | 5          |
| Ours (without Manhattan Loss)   | 149.86 | 5          |



**Figure 2: Qualitative comparison on RPLAN.**

## 4 RESULTS

Table 1 summarizes FID and model size: our single-stage model improves FID over HouseGAN++ but does not yet match two-stage GSDiff; removing the Manhattan alignment term severely degrades FID, confirming its utility. Numbers for HouseGAN++ and GSDiff are taken from [Hu et al. 2025]; unlike HouseGAN++, our method does not require bubble-diagram conditioning.

In Fig. 2, HouseGAN++ examples are reproduced from [Hu et al. 2025], and GSDiff outputs are generated in our environment using the authors' released model; HouseGAN++ often yields non-straight walls, GSDiff is more orthogonal yet shows occasional failures, and ours is typically clean and close to ground truth with rare failures (e.g., the rightmost Ours).

## 5 CONCLUSION

We introduced a single-stage graph beta diffusion framework with an unsupervised Manhattan alignment loss, achieving clear FID gains over HouseGAN++ while retaining a simpler pipeline. We will scale model capacity and conditioning to probe the FID-capacity trade-off against two-stage pipelines such as GSDiff.

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